

Four-year Undergraduate program (N.E.P)

Department of STATISTICS (Science Group A)

MJ 01. Semester -1Theory:

1. Major of dispersion
2. Moving average progressive average measure of Central tendency
3. Graphical representation of data Lawrence curve
4. Co relation
5. Regression.

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Semester 2- Major 2 credit 4 theory

1. Major of Association of attributes
2. Major of Central tendency
3. time series
4. index number
5. Statical quality control

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Semester second

Paper STS MJ 3 laboratory credit practical (75 + 25) marks List of Practical's

1. Distribution of population
2. Construct bar diagram
3. Double bar diagram
4. Rectangle square diagram
5. Graphical presentation of data
6. Find out A.M, median & mode
7. Find out mean deviation
8. Find out standard deviation
9. Find out variation
10. Coefficient of variation
11. Calculate percentile deviation.
12. Find out quartile deviation & percentile division
13. Construct Index number

14. Calculate cost of living index number
15. Estimate trend value
16. Method of list square
17. Method of list square
18. Feet a straight-line trend value
19. Calculate 4 yearly moving average
20. Draw control chart
21. Construct the defects control chart

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Semester 3- MJ 04

1. Nonparametric statistical method, power efficiency
2. Ordered statistics
3. Sign test, run Test median test
- 4th manner mars test Mann Whitney test 5 Finite difference.

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Semester 3 - MJ 05

1. Students t-test paired t test, z test, chi square test
2. Contingency table, test of independence, yules correction, F test
3. Analysis of variance
4. Index no
5. S.P.R.T

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Semester 4 MJ 06

1. Nonparametric statistical method, power efficiency
2. ordered statistics
3. Sign test, run Test median test, manner mars test Mann Whitney test
4. Demography, motility, fertility rate, life table
5. Finite difference.

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Semester 4 MJ7 contents

1. Fundamental of statistics, primary and secondary b data, kinds of statical investigation,
2. Economic statistics, index number
3. Elementary probability, bays theorem gauss Markov theorem
4. Correlation method, probable error
, multiple correlation, partial correlation
5. Regression analysis, Scatter diagram confidence limit, regression line function.

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Semester 4 MJ 08

List of practical (75 + 25) marks

1. Find out SD
2. Find out variance
3. Calculate median
4. Calculate mean for frequency distribution
5. Calculate missing frequency
6. Calculate mean deviation
7. Find out regression
8. Find Quartile deviation.
9. Construct bar diagram
10. Construct Histogram

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Semester 5 MJ 9 contents

1. statical technique in quality control, pie chart, Sigma chart, R chart
2. SQC, p chart c chart, modified control, acceptance sampling plan,
3. consumer risk, acceptance quality level
4. collection of data, method of collection of data, questionnaire, census and sample survey.
5. concept classification, function limitation, moving average, progressive average, composite average, pooled mean, Major of Central tendency

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Semester 5 - Major 10

1. Methods of measuring, linear trend, seasonal variation, link relative method, cycling variation, measurement of in regular variation.
2. Mean absolute deviation, coefficient of variation moment, different missing.
3. time revival test, factor reversal test, consumer, price index, net National product, cause National product, industrial product.
4. sampling methods, population, Finite, infinite, real, random sampling, non-sampling error, parameter, sample size, sample mean, variance.
5. Design of experiments stratified random sampling, allocation optimum, Neyman, systematic sampling, two stage sampling.

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Semester 5 MJ 11

1. The theory of estimation, properties, mean square error, crammer row in quality, normal estimators, efficiency.
2. testing parametric hypothesis, power of test, best critical reason.
3. degree of freedom, non-randomized test, optimum test, likelihood ratio test.
4. sky squad test yet correction attest analysis of variance-based test.
5. S.P.R.T Lawrence curve

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Semester 6 MJ 12 contents

- 1 Cost of living index number.
2. Discreet distribution.
3. continuous distribution.
4. random variable, property of distribution function, probability mass function, Discrete distribution function.
5. Distribution function, marginal distribution function, joint density function, stochastic independence.

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Major 13 - Semester 6

1. Mathematical expectation, covariance variance of linear combination, random variable.
2. Probability generating function, multivariate characteristics, weak law of large number.
3. Bernoulli distribution, binomial Distribution, moment.
4. Negative binomial Distribution moment generating function of geometric distribution lack of memory hypergeometric distribution.
5. Theoretical discrete distribution, hypergeometric distribution.

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Semester 6 Major 14

1. Theoretical continuous distribution
2. Principle of least square.
3. Correlation ratio, inter class correlation, Bivariate normal distribution.
4. Marginal distribution, multiple, partial correlation, property of multiple correlation coefficient of partial correlation.
5. Theory of attributes.

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MJ 15 sem 6 list of practical

1. Find the coefficient of correlation.
2. Calculation of Pearson's coefficient of correlation
3. Find rank correlation.
4. By method of concurrent deviation calculate correlation.
5. Regression equation
6. Estimate regression equation line.
7. Calculate regression equation.
8. Find the value of correlation coefficient.
- 9 Find the regression equation of Y on X.
10. Find the first four moments.
- 11 Find the first four Central moments.
12. Calculate coefficient of SKEWNESS.
13. Find standard deviation.
14. Find quartile deviation.
15. Find mean deviation.

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semester 7 MJ 16

1. Sampling.
2. large sample tests.
3. Exact sampling distribution.
4. T, F, and z distribution.
5. Non parametric methods and sequential analysis.

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Semester 7 MJ 17

1. Basic experimental design.
2. Factorial experiments.
3. Vital statistics.
4. Population statistics.
5. Statistical technique in quality control.

Department of statistics MJ 18 semester 7 contents

1. Interpolation & extrapolation
2. Distribution
3. Demography
4. Sample survey.
5. Design of experiment.

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MJ 19 semester 7 List of practical

1. Calculation of index number.
2. Using sample average index number.
3. Construct fisher's index number.
4. Find cost of living index number.
5. For probability distribution calculate mean and variance.
6. Find mathematical expectation of number.
7. Calculate mean deviation and coefficient of mean deviation.
8. Calculate mean deviation about median.
9. Find standard deviation.
10. Find standard deviation and its coefficient.
11. Fit a poisson distribution.
12. Calculate chi square.
13. Calculate chi square of given distribution.
14. Fit parabolas of second degree.
15. Calculate sample mean and standard deviation.

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Semester 8 MJ 20

Contents theory of sample survey questionnaire basis of field work. department of statistics.

Department of statistics (science) advanced Major 1

1. Classification, Tabulation & frequency distribution
2. Sampling methods.
3. Random variable, mathematical expectation and probability distribution.
- .4. Theory of Estimation.
5. Continuous distribution, sampling distribution.
6. Basic experimental design.
index number.

Department of statistics Advance Major 2 Semester - 8:

1. Bivariate random variable and distribution.
2. Testing parametric hypothesizes.
3. Nonparametric statistical methods.
4. Regression and Correlation methods.
5. Measure of an Association of attributes.
6. Interpolation and extra polation.
7. Vital and population statistics.

Advance Major 3 List of Practical Semester - 8

1. Calculation of index number.
2. Using sample average index number.
3. Construct fisher's index number.
4. Find cost of living index number.
5. For probability distribution calculate mean and variance.
6. Find mathematical expectation of number.
7. Calculate mean deviation and coefficient of mean deviation.
8. Calculate mean deviation about median.
9. Find standard deviation.
10. Find standard deviation and its coefficient.
11. Fit a poisson distribution.
12. Calculate chi square.
13. Calculate chi square of given distribution.

14 Fit parabola of second degree

15. Calculate sample mean and standard deviation.

16. Construct life.

17. Construct Graphical & Diagrammatic representation of data.

18. Problem based on Finite difference table.

19. Find out interpolation and extrapolation.

Department of statistics (Science group A) Minor-I

Theory: 75 (15 + 60) marks

1. Graphical representation of data.
2. Correlation & Regression
3. Time series
4. Index number
5. Distribution
6. Statistical quality control
7. Testing of HYPOTHESIS.

Practical: 25 marks

1. Construct bar diagram
2. Double bar diagram
3. Rectangle square diagram
4. Graphical representation of data
5. Find out arithmetic mean median mode
6. Find out mean deviation
7. Find out the standard division.

Department of statistics (Science group A) Minor-II

Theory: 75 (15 + 60) marks

1. Interpolation & Extrapolation
2. Discrete distribution.
3. Continuous distribution.
4. Sample survey
5. Design of experiment.

Practical: 25 marks

1. Find out variation
2. Coefficient of variation
3. Calculate percentile division
4. Find out quartile deviation
5. Construct Index number

Department of statistics (Science group A) Minor-III

Theory: 75 (15 + 60) marks

1. Theory of estimation.
2. Sampling methods.
3. Frequency distribution.
4. Random variable, Mathematical expectation and probability distribution.
5. Bivariate random variable and distribution.

Practical: 25 marks

1. Draw control chart
2. Fit a straight-line trend value
3. Find out correlation
4. Find out regression line
5. Method of list square.

Department of statistics (Science group A) Minor-IV

Theory: 75 (15 + 60) marks

1. Elementary probability.
2. Testing parametric hypothesis
3. Nonparametric statistical methods.
4. Regression and correlation methods
5. Vital and population statistics.

Practical: 25 marks

1. Calculate cost of living index number
2. Find out standard deviation
3. Find out variance
4. Construct histogram
5. Calculate Chi square

NEP-2020, FYDUGP

STATISTICS (science group A) Multi-disciplinary MCD

1. Statistical quality control (SQC).
2. TESTING of hypothesis.
3. TIME series.
4. Graphical representation of data.
5. Measure of DISPERSION.

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Suggested reading books :

1. Principle of statistics - Dr S.M. Shukla and Dr. S.P Sahay
2. Applied in business statistics Dr. K.N.Gupta
3. Fundamental of mathematical statistics S.C. Gupta and V.K. Kapoor Sultan Chand and sons' publication
4. Statistics Dr B.N. Gupta
5. Statistics theory and practical by S.P.Singh
6. Fundamental of statistics for one and two Volumes by Goon and Gupta.

Latest editions of Text. books may be used.